

ANN-calibrated spatial masking in PING

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Abstract

A pyramidal–interneuron gamma (PING) network trained on clean grayscale handwritten digits encounters permanent removal of foreground pixels. An architecture-matched artificial neural network (ANN) calibrates when the damaged stimulus itself ceases to support classification. Under identical spatial masks, PING outperforms the ANN at intermediate deletion but reaches chance sooner: at retention $q = 0.02$, PING is at 10.7% while the ANN remains at 21.7%. The calibrated range now bounds a follow-up training experiment with variable evidence.

Methods

The dataset is MNIST, a ten-class collection of grayscale handwritten digits. The standard corpus is split into 56000 training and 14000 held-out examples. The protocol has two ordered parts:

1. **Calibrate foreground deletion with an ANN.** Train 3 seeds of a non-spiking classifier with 784 inputs, one rectified-linear-unit (ReLU) hidden layer of 1024 units (matching the PING excitatory population), and 10 outputs. Training uses 15 epochs, batches of 256, and learning rate 0.001. Binarize each held-out image at intensity 0, then retain every foreground pixel independently with probability q . Here $q = 1$ leaves the foreground intact and $q = 0$ removes it. Evaluate 10 mask draws per image at every registered q .
2. **Compare spatial masking directly.** Select 100 fixed held-out images. Present the identical binarized images and Bernoulli masks to every ANN and PING seed. PING uses a maximum encoding rate of 25 Hz and its trained presentation duration. This paired subset makes differences attributable to the classifiers rather than different corruptions.

Results

The ANN needs almost complete foreground removal to reach chance

The architecture-matched ANN remains above chance until foreground retention falls to $q = 0.005$, fewer than one visible foreground pixel per image on average.

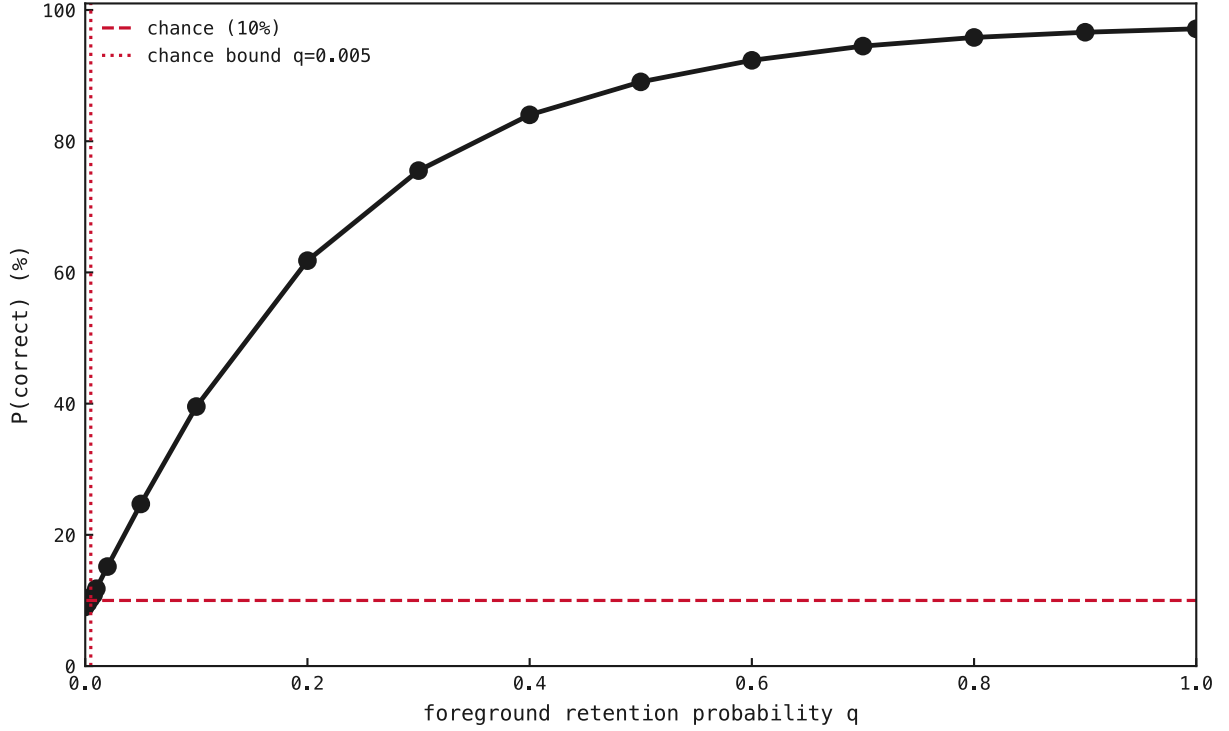


Figure 1: Held-out ANN classification accuracy as foreground evidence is removed. The horizontal axis is foreground-retention probability q (dimensionless); the vertical axis is the probability of a correct classification, in percent. Points are means across the architecture-matched ANN seeds and the band is one standard error across seeds. The dashed red line is ten-class chance performance; the dotted line marks the chance-region bound read from the run. Accuracy stays above chance until almost every foreground pixel is absent.

PING is competitive at intermediate masking but reaches chance sooner

The matched comparison does not have one uniformly better classifier. With 29.5 visible pixels on average ($q = 0.2$), PING reaches 68.7% against the ANN's 60.7%. They coincide near 45% at $q = 0.1$. PING still leads at $q = 0.05$, then reaches chance at $q = 0.02$ while the ANN remains above it. The PING transition is steeper, not uniformly worse.

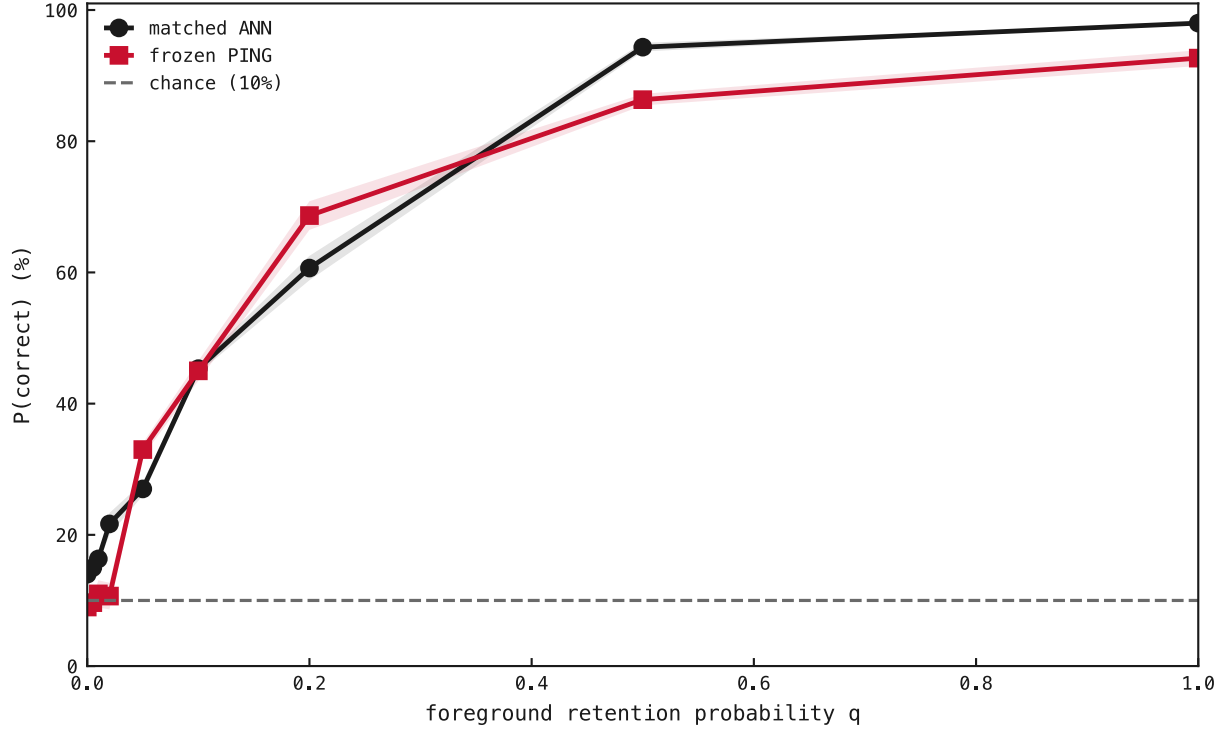


Figure 2: Architecture-matched ANN and frozen PING classification under identical spatial deletion. The horizontal axis is foreground-retention probability q (dimensionless); the vertical axis is the probability of a correct classification, in percent. Black circles denote ANN and red squares PING, so marker shape also identifies each classifier in grayscale. Each point uses 100 fixed held-out examples and the same Bernoulli foreground masks; uncertainty is one standard error across trained seeds. PING runs at 25 Hz for 200 ms. PING is competitive at intermediate retention but crosses into its chance region before the ANN.

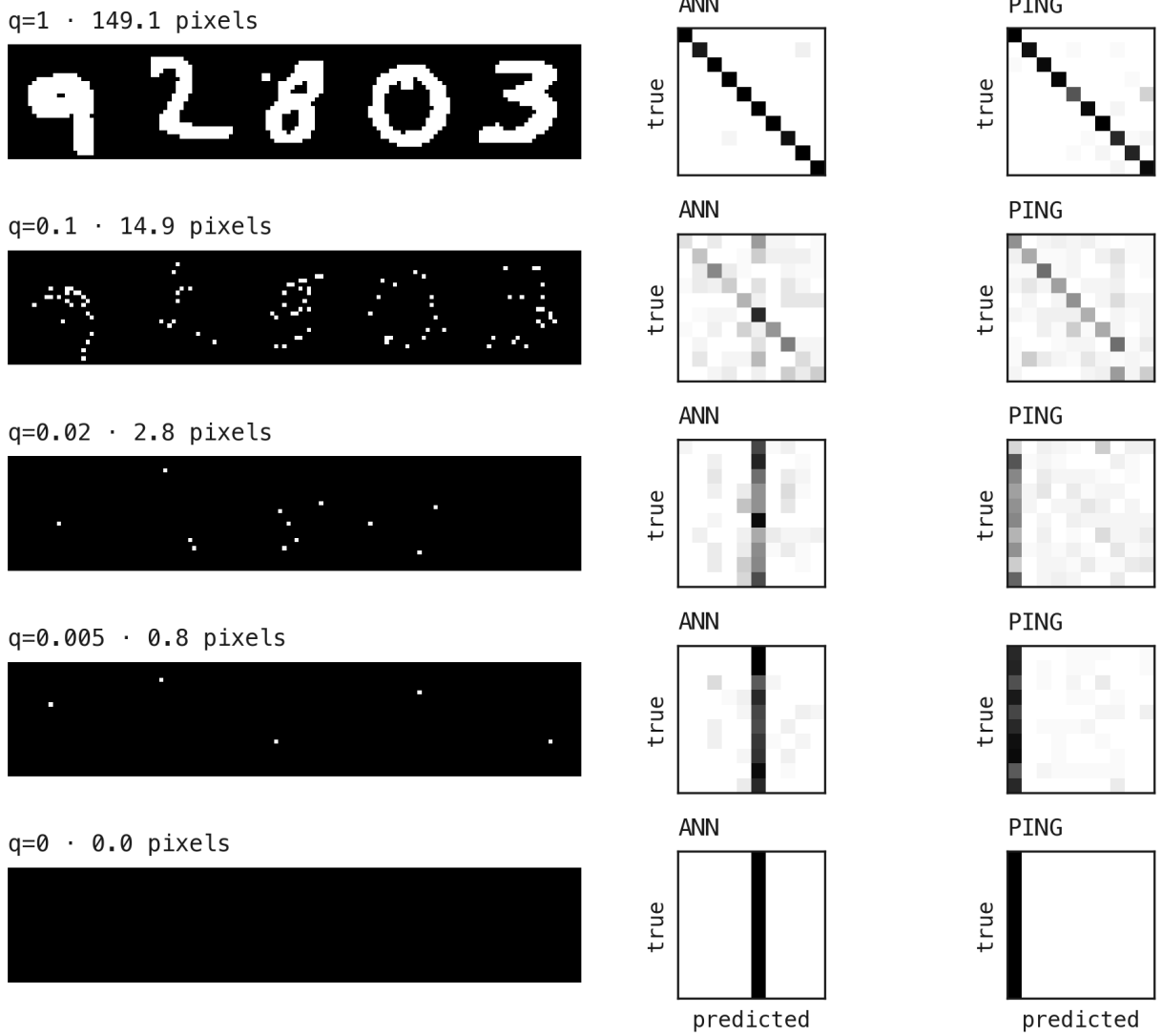


Figure 3: Stimulus and error structure across the matched masking curve. Rows progress from intact input through intermediate masking, the PING and ANN chance regions, and the blank control; the left panel in each row shows five example digits and reports q plus the mean number of visible foreground pixels. The middle and right panels are row-normalized ANN and PING confusion matrices: rows are true classes and columns predicted classes. The blank-input columns reflect each model’s constant prediction and the class prevalence in this fixed subset, not evidence extracted from a blank image.

Caveat. Binarization maps every nonzero antialiased MNIST pixel to full intensity. At $q = 1$ that drives PING more strongly than the original grayscale image at the same encoding-rate ceiling. This curve therefore measures robustness to permanent spatial deletion from a binarized stimulus; it is not a second measurement of grayscale contrast.

The chance bounds define the spatial training range

The ANN reaches chance near $q = 0.005$; matched PING by $q = 0.02$. These bounds set the range for variable-evidence training. Encoding-rate reduction and spatial masking remain distinct because they remove different evidence. The interval between the PING and ANN chance bounds identifies extreme corruptions where some class information remains usable by

the ANN but not by PING; at moderate masking, PING is competitive rather than uniformly fragile. Encoding-rate viability is treated separately in exp048.